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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 31st August Evening Shift

JEE Main Question Paper with Answer Key

31 Aug 2021	90	21	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	21 questions
Mathematics	20 questions
Physics	28 questions
Out of syllabus	21 questions
Answer key	at the end

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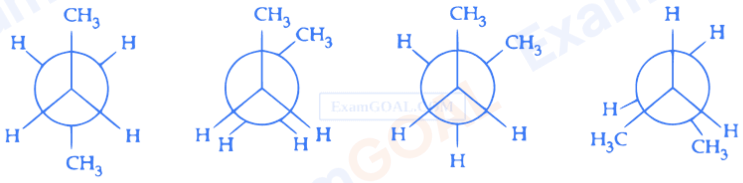
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2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Chemistry 21 questions

Q1. MCQ (Single Correct Answer)

Open solution

Arrange the following conformational isomers of n-butane in order of their increasing potential energy :



I II III IV

(A) II < III < IV < I

(B) I < IV < III < II

(C) II < IV < III < I

(D) I < III < IV < II

Q2. MCQ (Single Correct Answer)

Open solution

The Eu^{2+} ion is a strong reducing agent in spite of its ground state electronic configuration (outermost) : [Atomic number of Eu = 63]

(A) $4f^7 6s^2$

(B) $4f^6$

(C) $4f^7$

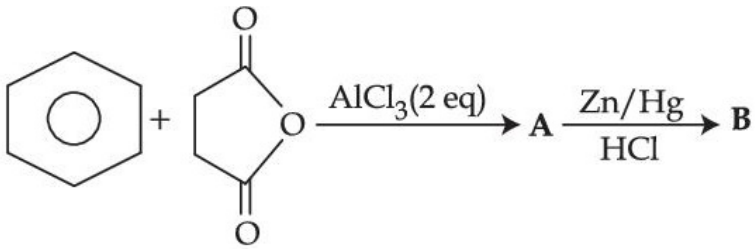
(D) $4f^6 6s^2$

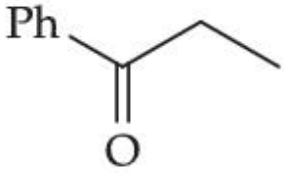
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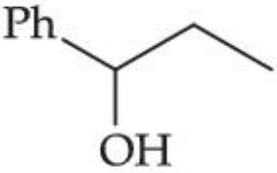
Q3. MCQ (Single Correct Answer)

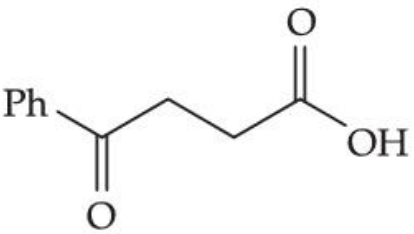
Open solution

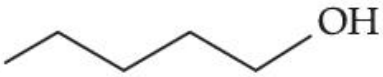
The structures A and B formed in the following reaction are : [Ph = $-C_6H_5$]

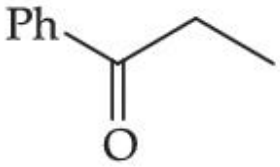


(A) A = 

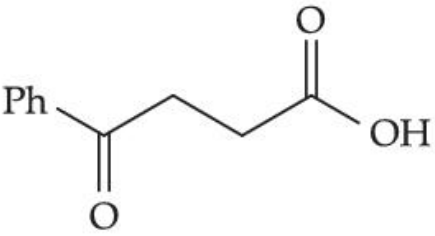
, B = 

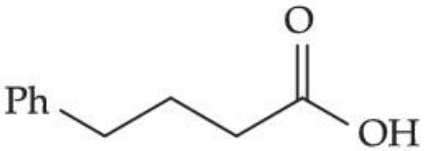
(B) A = 

, B = Ph 

(C) A = 

, B = 

(D) A = 

, B = 

Q4. MCQ (Single Correct Answer)

Open solution

In which one of the following sets all species show disproportionation reaction?

(A) ClO_2^-, F_2, MnO_4^- and $Cr_2O_7^{2-}$

(B) $Cr_2O_7^{2-}$, MnO_4^-, ClO_2^- and Cl_2

(C) MnO_4^-, ClO_2^-, Cl_2 and Mn^{3+}

(D) ClO_4^-, MnO_4^-, ClO_2^- and F_2

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Q5. MCQ (Single Correct Answer)

Match List - I with List - II

Open solution

	List - I (Parameter)		List - II (Unit)
(a)	Cell constant	(i)	$S\,cm^2mol^{-1}$
(b)	Molar conductivity	(ii)	Dimensionless
(c)	Conductivity	(iii)	m^{-1}
(d)	Degree of dissociation of electrolyte	(iv)	$\Omega^{-1}m^{-1}$

Choose the most appropriate answer from the options given below :

(A) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(B) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(C) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

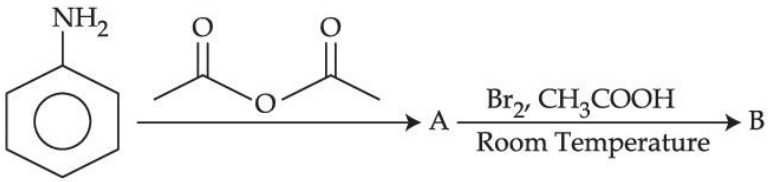
(D) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

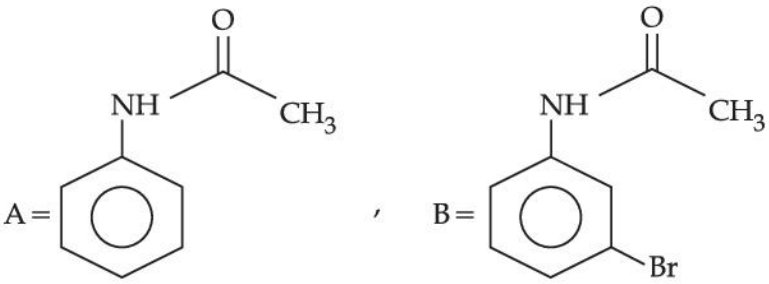
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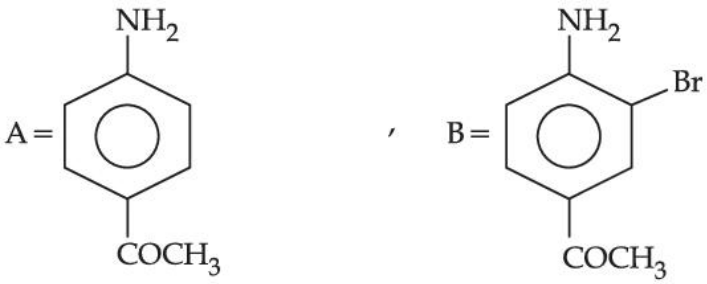
Q6. MCQ (Single Correct Answer)

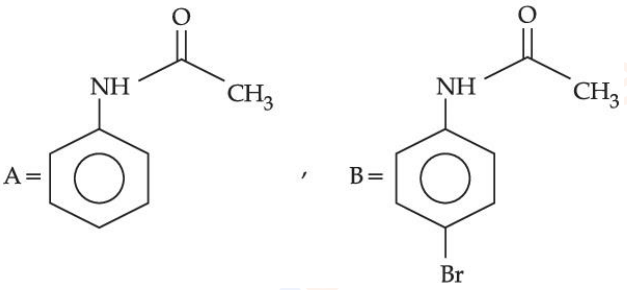
Open solution

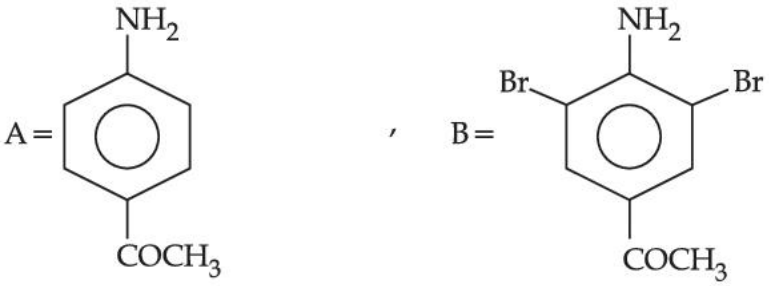
The major products A and B formed in the following reaction sequence are :



(A) 

(B) 

(C) 

(D) 

Q7. MCQ (Single Correct Answer)

Open solution

Which of the following is not an example of fibrous protein?

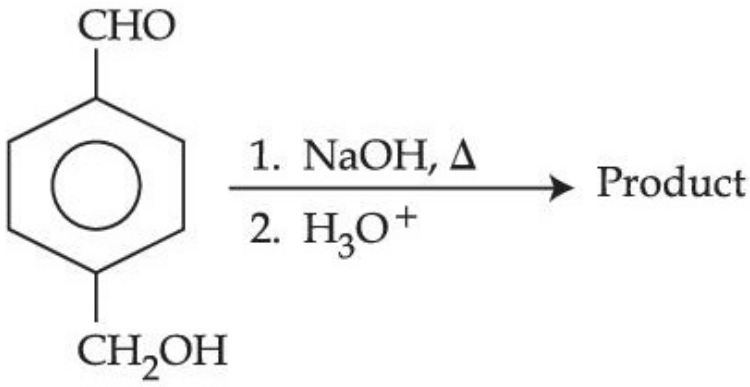
(A) Keratin (B) Albumin (C) Collagen (D) Myosin

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Q8. MCQ (Single Correct Answer)

Open solution

For the reaction given below :-



The compound which is not formed as a product in the reaction is a :

(A) compound with both alcohol and acid functional groups

(B) monocarboxylic acid

(C) dicarboxylic acid

(D) diol

Q9. MCQ (Single Correct Answer)

Open solution

Spin only magnetic moment in BM of $[Fe(CO)_4(C_2O_4)]^+$ is :

(A) 5.92 (B) 0 (C) 1 (D) 1.73

Q10. MCQ (Single Correct Answer)

Open solution

The incorrect expression among the following is :

(A) $\frac{\Delta G_{System}}{\Delta S_{Total}} = -T$ (at constant P)

(B) $K = \frac{\Delta H^\circ - T\Delta S^\circ}{RT}$

(C) $K = e^{-\Delta G^\circ / RT}$

(D) For isothermal process $w_{reversible} = -nRT \ln \frac{V_f}{V_i}$

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Q11. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II :

	List - I (Metal Ion)		List - II (Group of Qualitative analysis)
(a)	Mn^{2+}	(i)	Group - III
(b)	As^{3+}	(ii)	Group - IIA
(c)	Cu^{2+}	(iii)	Group - IV
(d)	Al^{3+}	(iv)	Group - IIB

Choose the most appropriate answer from the options given below :

(A) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(B) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(C) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)

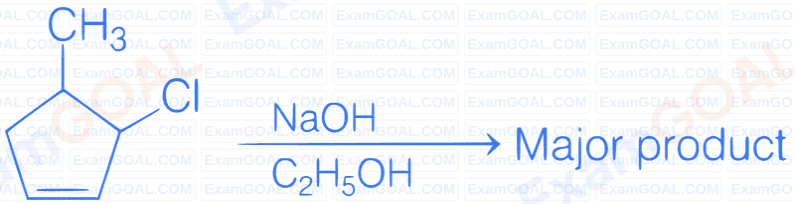
(D) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

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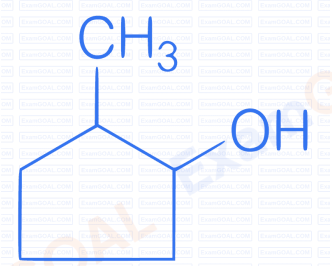
Q12. MCQ (Single Correct Answer)

Open solution

The major product of the following reaction is :



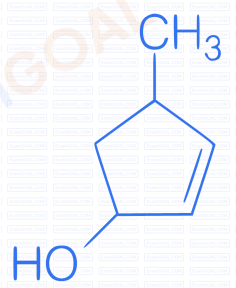
(A)



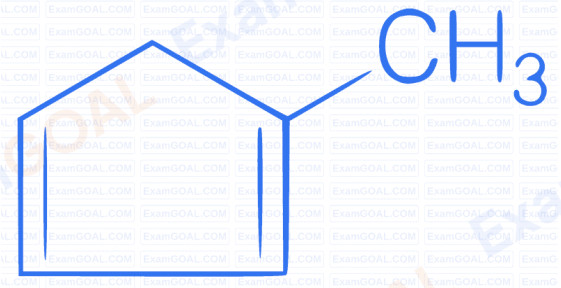
(B)



(C)



(D)

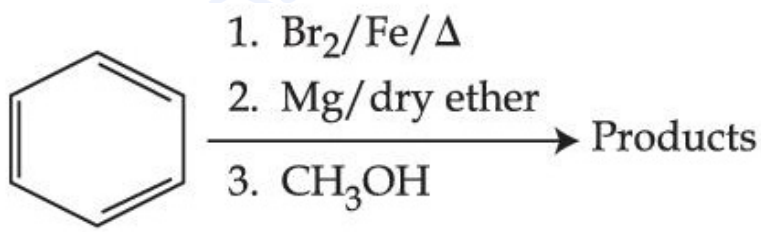


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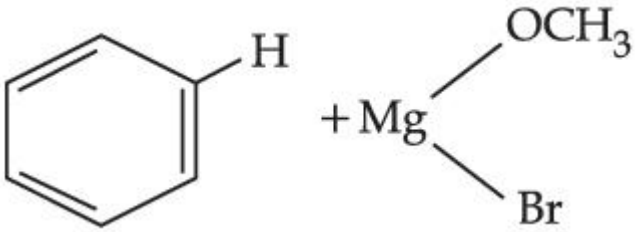
Q13. MCQ (Single Correct Answer)

Open solution

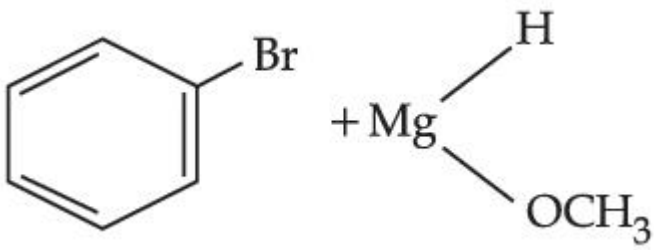
For the following sequence of reactions, the correct products are :



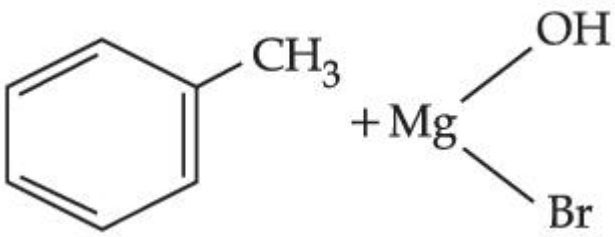
(A)



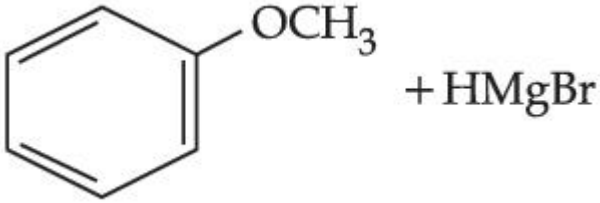
(B)



(C)



(D)

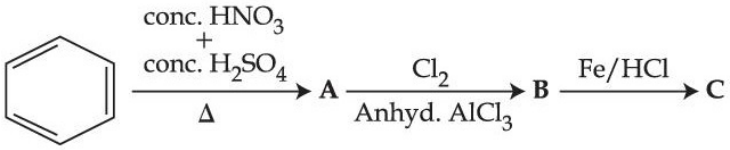


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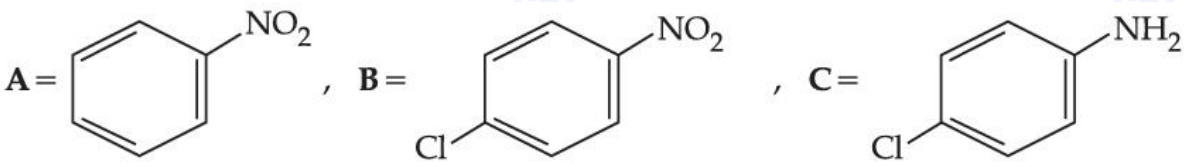
Q14. MCQ (Single Correct Answer)

Open solution

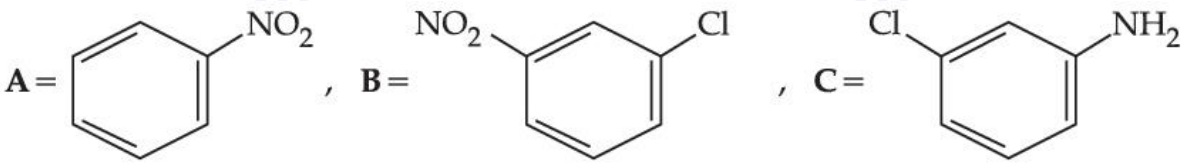
Identify correct A, B and C in the reaction sequence given below :



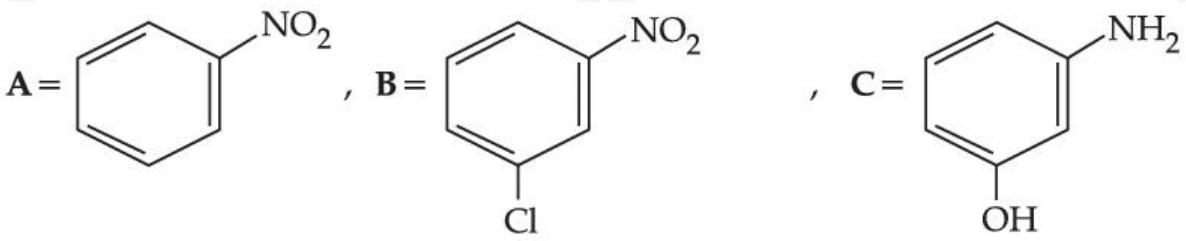
(A)



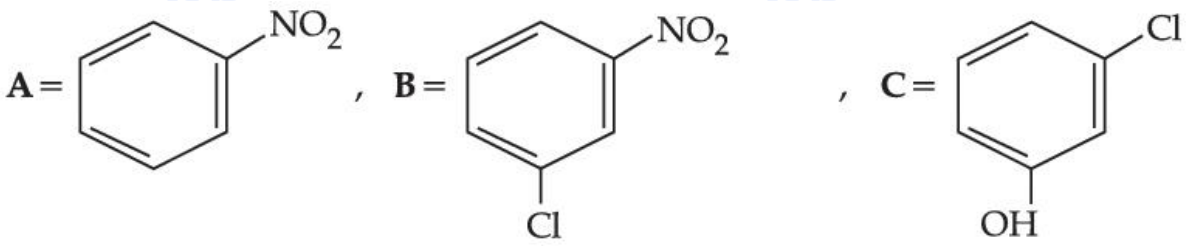
(B)



(C)



(D)



Q15. Numerical

Open solution

1.22 g of an organic acid is separately dissolved in 100 g of benzene (K_b = 2.6 K kg mol^{-1}) and 100 g of acetone (K_b = 1.7 K kg mol^{-1}). The acid is known to dimerize in benzene but remain as a monomer in acetone. The boiling point of the solution in acetone increases by 0.17°C. The increase in boiling point of solution in benzene in °C is $x \times 10^{-2}$. The value of x is _____. (Nearest integer) [Atomic mass : C = 12.0, H = 1.0, O =16.0]

Q16. Numerical

Open solution

The value of magnetic quantum number of the outermost electron of Zn^+ ion is _____.

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Q17. Numerical

The pH of a solution obtained by mixing 50 mL of 1 M HCl and 30 mL of 1 M NaOH is x $\times$ 10^{-4}. The value of x is _____. (Nearest integer) [log 2.5 = 0.3979]

Open solution

Q18. Numerical

For the reaction A $\rightarrow$ B, the rate constant k(in s^{-1}) is given by

$$\log_{10}k = 20.35 - \frac{(2.47\times10^3)}{T}$$

The energy of activation in kJ mol^{-1} is _____. (Nearest integer) [Given : R = 8.314 J K^{-1} mol $^{-1}$]

Open solution

Q19. Numerical

Sodium oxide reacts with water to produce sodium hydroxide. 20.0 g of sodium oxide is dissolved in 500 mL of water. Neglecting the change in volume, the concentration of the resulting NaOH solution is _____ $\times$ 10^{-1} M. (Nearest integer) [Atomic mass : Na = 23.0, O = 16.0, H = 1.0]

Open solution

Q20. Numerical

According to molecular orbital theory, the number of unpaired electron(s) in O_2^{2-} is :

Open solution

Q21. Numerical

The transformation occurring in Duma's method is given below :

$$C_2H_7N + \left(2x + \tfrac{y}{2}\right)CuO \rightarrow xCO_2 + \tfrac{y}{2}H_2O + \tfrac{z}{2}N_2 + \left(2x + \tfrac{y}{2}\right)Cu$$

The value of y is _____. (Integer answer)

Open solution

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Mathematics 20 questions

Q22. MCQ (Single Correct Answer)

If $\alpha + \beta + \gamma = 2\pi$, then the system of equations

$x + (\cos\gamma)y + (\cos\beta)z = 0$

$(\cos\gamma)x + y + (\cos\alpha)z = 0$

$(\cos\beta)x + (\cos\alpha)y + z = 0$

has :

(A) no solution
(B) infinitely many solution
(C) exactly two solutions
(D) a unique solution

Open solution

Q23. MCQ (Single Correct Answer)

The domain of the function

$f(x) = \sin^{-1}\left(\frac{3x^2+x-1}{(x-1)^2}\right) + \cos^{-1}\left(\frac{x-1}{x+1}\right)$ is :

(A) $\left[0, \frac{1}{4}\right]$
(B) $[-2, 0] \cup \left[\frac{1}{4}, \frac{1}{2}\right]$
(C) $\left[\frac{1}{4}, \frac{1}{2}\right] \cup \{0\}$
(D) $\left[0, \frac{1}{2}\right]$

Open solution

Q24. MCQ (Single Correct Answer)

Let S = {1, 2, 3, 4, 5, 6}. Then the probability that a randomly chosen onto function g from S to S satisfies g(3) = 2g(1) is :

(A) $\frac{1}{10}$
(B) $\frac{1}{15}$
(C) $\frac{1}{5}$
(D) $\frac{1}{30}$

Open solution

Q25. MCQ (Single Correct Answer)

Let f : N $\to$ N be a function such that f(m + n) = f(m) + f(n) for every m, n$\in$N. If f(6) = 18, then f(2) . f(3) is equal to :

(A) 6
(B) 54
(C) 18
(D) 36

Open solution

Q26. MCQ (Single Correct Answer)

If $\alpha = \lim\limits_{x\to\frac{\pi}{4}} \frac{\tan^3 x - \tan x}{\cos\left(x+\frac{\pi}{4}\right)}$ and $\beta = \lim\limits_{x\to 0} (\cos x)^{\cot x}$ are the roots of the equation, $ax^2 + bx - 4 = 0$, then the ordered pair (a, b) is :

(A) $(1, -3)$
(B) $(-1, 3)$
(C) $(-1, -3)$
(D) $(1, 3)$

Open solution

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Q27. MCQ (Single Correct Answer)

The locus of mid-points of the line segments joining $(-3, -5)$ and the points on the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ is :

(A) $9x^2 + 4y^2 + 18x + 8y + 145 = 0$

(B) $36x^2 + 16y^2 + 90x + 56y + 145 = 0$

(C) $36x^2 + 16y^2 + 108x + 80y + 145 = 0$

(D) $36x^2 + 16y^2 + 72x + 32y + 145 = 0$

Open solution

Q28. MCQ (Single Correct Answer)

If $\frac{dy}{dx} = \frac{2^x y + 2^y . 2^x}{2^x + 2^{x+y}\log_e 2}$, y(0) = 0, then for y = 1, the value of x lies in the interval :

(A) (1, 2)

(B) $\left(\frac{1}{2}, 1\right]$

(C) (2, 3)

(D) $\left(0, \frac{1}{2}\right]$

Open solution

Q29. MCQ (Single Correct Answer)

If $y\frac{dy}{dx} = x\left[\frac{y^2}{x^2} + \frac{\phi\left(\frac{y^2}{x^2}\right)}{\phi'\left(\frac{y^2}{x^2}\right)}\right]$, x > 0, ϕ > 0, and y(1) = -1, then $\phi\left(\frac{y^2}{4}\right)$ is equal to :

(A) 4 ϕ (2)

(B) 4ϕ (1)

(C) 2 ϕ (1)

(D) ϕ (1)

Open solution

Q30. MCQ (Single Correct Answer)

The sum of the roots of the equation

$x + 1 - 2\log_2(3 + 2^x) + 2\log_4(10 - 2^{-x}) = 0$, is :

(A) $\log_2 14$

(B) $\log_2 11$

(C) $\log_2 12$

(D) $\log_2 13$

Open solution

Q31. MCQ (Single Correct Answer)

If z is a complex number such that $\frac{z-i}{z-1}$ is purely imaginary, then the minimum value of | z $-$ (3 + 3i) | is :

(A) $2\sqrt{2} - 1$

(B) $3\sqrt{2}$

(C) $6\sqrt{2}$

(D) $2\sqrt{2}$

Open solution

Q32. MCQ (Single Correct Answer)

Let $a_1, a_2, a_3, \ldots$ be an A.P. If $\frac{a_1 + a_2 + \ldots + a_{10}}{a_1 + a_2 + \ldots + a_p} = \frac{100}{p^2}$, p $\neq$ 10, then $\frac{a_{11}}{a_{10}}$ is equal to :

(A) $\frac{19}{21}$

(B) $\frac{100}{121}$

(C) $\frac{21}{19}$

(D) $\frac{121}{100}$

Open solution

Q33. MCQ (Single Correct Answer)

Let A be the set of all points (α, β) such that the area of triangle formed by the points (5, 6), (3, 2) and (α, β) is 12 square units. Then the least possible length of a line segment joining the origin to a point in A, is :

(A) $\frac{4}{\sqrt{5}}$

(B) $\frac{16}{\sqrt{5}}$

(C) $\frac{8}{\sqrt{5}}$

(D) $\frac{12}{\sqrt{5}}$

Open solution

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Q34. MCQ (Single Correct Answer)

If [x] is the greatest integer $\leq$ x, then

$\pi^2 \int_0^2 \left(\sin \frac{\pi x}{2}\right)(x-[x])^{[x]} dx$ is equal to :

(A) $2(\pi - 1)$ (B) $4(\pi - 1)$ (C) $4(\pi + 1)$ (D) $2(\pi + 1)$

Open solution

Q35. MCQ (Single Correct Answer)

The mean and variance of 7 observations are 8 and 16 respectively. If two observations are 6 and 8, then the variance of the remaining 5 observations is :

(A) $\frac{92}{5}$ (B) $\frac{134}{5}$ (C) $\frac{536}{25}$ (D) $\frac{112}{5}$

Open solution

Q36. Numerical

If the coefficient of a^7b^8 in the expansion of $(a + 2b + 4ab)^{10}$ is $K.2^{16}$, then K is equal to _____.

Open solution

Q37. Numerical

The number of 4-digit numbers which are neither multiple of 7 nor multiple of 3 is _____.

Open solution

Q38. Numerical

If $\int \frac{\sin x}{\sin^3 x + \cos^3 x} dx =$

$\alpha \log_e|1 + \tan x| + \beta \log_e|1 - \tan x + \tan^2 x| + \gamma \tan^{-1}\left(\frac{2\tan x - 1}{\sqrt{3}}\right) + C$, where C is constant of integration, then the value of $18(\alpha + \beta + \gamma^2)$ is _____.

Open solution

Q39. Numerical

The number of elements in the set $\left\{A = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} : a, b, d \in \{-1, 0, 1\} \text{ and } (I - A)^3 = I - A^3\right\}$, where I is 2×2 identity matrix, is :

Open solution

Q40. Numerical

If the line y = mx bisects the area enclosed by the lines x = 0, y = 0, x = $\frac{3}{2}$ and the curve $y = 1 + 4x - x^2$, then 12 m is equal to _____.

Open solution

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Q41. Numerical

Let f(x) be a cubic polynomial with f(1) = −10, f(−1) = 6, and has a local minima at x = 1, and f'(x) has a local minima at x = −1. Then f(3) is equal to _____.

Open solution

Physics 28 questions

Q42. MCQ (Single Correct Answer)

Four identical hollow cylindrical columns of mild steel support a big structure of mass 50×10^3 kg. The inner and outer radii of each column are 50 cm and 100 cm respectively. Assuming uniform local distribution, calculate the compression strain of each column. [Use Y = 2.0×10^{11} Pa, g = 9.8 m/s^2]

(A) 3.60×10^{-8} (B) 2.60×10^{-7} (C) 1.87×10^{-3} (D) 7.07×10^{-4}

Open solution

Q43. MCQ (Single Correct Answer)

A current of 1.5 A is flowing through a triangle, of side 9 cm each. The magnetic field at the centroid of the triangle is :

(Assume that the current is flowing in the clockwise direction.)

(A) 3×10^{-7} T, outside the plane of triangle

(B) $2\sqrt{3} \times 10^{-7}$ T, outside the plane of triangle

(C) $2\sqrt{3} \times 10^{-5}$ T, inside the plane of triangle

(D) 3×10^{-5} T, inside the plane of triangle

Open solution

Q44. MCQ (Single Correct Answer)

A system consists of two identical spheres each of mass 1.5 kg and radius 50 cm at the ends of a light rod. The distance between the centres of the two spheres is 5 m. What will be the moment of inertia of the system about an axis perpendicular to the rod passing through its midpoint?

(A) 18.75 kgm^2 (B) 1.905×10^5 kgm^2

(C) 19.05 kgm^2 (D) 1.875×10^5 kgm^2

Open solution

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Q45. MCQ (Single Correct Answer)

Open solution

Statement I :

Two forces $\left(\vec{P}+\vec{Q}\right)$ and $\left(\vec{P}-\vec{Q}\right)$ where $\vec{P}\perp\vec{Q}$, when act at an angle θ_1 to each other, the magnitude of their resultant is $\sqrt{3(P^2+Q^2)}$, when they act at an angle θ_2, the magnitude of their resultant becomes $\sqrt{2(P^2+Q^2)}$. This is possible only when $\theta_1 < \theta_2$.

Statement II :

In the situation given above.

θ_1 = 60° and θ_2 = 90°

In the light of the above statements, choose the most appropriate answer from the options given below :-

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are false.

Q46. MCQ (Single Correct Answer)

Open solution

A free electron of 2.6 eV energy collides with a H^+ ion. This results in the formation of a hydrogen atom in the first excited state and a photon is released. Find the frequency of the emitted photon. (h = 6.6×10^{-34} Js)

(A) 1.45×10^{16} MHz

(B) 0.19×10^{15} MHz

(C) 1.45×10^{9} MHz

(D) 9.0×10^{27} MHz

Q47. MCQ (Single Correct Answer)

Open solution

Two thin metallic spherical shells of radii r_1 and r_2 ($r_1 < r_2$) are placed with their centres coinciding. A material of thermal conductivity K is filled in the space between the shells. The inner shell is maintained at temperature θ_1 and the outer shell at temperature $\theta_2(\theta_1 < \theta_2)$. The rate at which heat flows radially through the material is :-

(A) $\frac{4\pi K r_1 r_2(\theta_2-\theta_1)}{r_2-r_1}$

(B) $\frac{\pi r_1 r_2(\theta_2-\theta_1)}{r_2-r_1}$

(C) $\frac{K(\theta_2-\theta_1)}{r_2-r_1}$

(D) $\frac{K(\theta_2-\theta_1)(r_2-r_1)}{4\pi r_1 r_2}$

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Q48. MCQ (Single Correct Answer)

Open solution

Consider two separate ideal gases of electrons and protons having same number of particles. The temperature of both the gases is the same. The ratio of the uncertainty in determining the position of an electron to that of a proton is proportional to :-

(A) $\left(\frac{m_p}{m_e}\right)^{3/2}$

(B) $\sqrt{\frac{m_e}{m_p}}$

(C) $\sqrt{\frac{m_p}{m_e}}$

(D) $\frac{m_p}{m_e}$

Q49. MCQ (Single Correct Answer)

Open solution

A bob of mass 'm' suspended by a thread of length l undergoes simple harmonic oscillations with time period T. If the bob is immersed in a liquid that has density $\frac{1}{4}$ times that of the bob and the length of the thread is increased by 1/3rd of the original length, then the time period of the simple harmonic oscillations will be :-

(A) T

(B) $\frac{3}{2}$T

(C) $\frac{3}{4}$T

(D) $\frac{4}{3}$T

Q50. MCQ (Single Correct Answer)

Open solution

Statement : 1

If three forces $\vec{F_1}, \vec{F_2}$ and $\vec{F_3}$ are represented by three sides of a triangle and $\vec{F_1} + \vec{F_2} = -\vec{F_3}$, then these three forces are concurrent forces and satisfy the condition for equilibrium.

Statement : 2

A triangle made up of three forces $\vec{F_1}, \vec{F_2}$ and $\vec{F_3}$ as its sides taken in the same order, satisfy the condition for translatory equilibrium.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement - I is false but Statement - II is true

(B) Statement - I is true but Statement - II is false

(C) Both Statement-I and Statement-II are false

(D) Both Statement-I and Statement-II are true

Q51. MCQ (Single Correct Answer)

Open solution

If velocity [V], time [T] and force [F] are chosen as the base quantities, the dimensions of the mass will be :

(A) $[FT^{-1}\,V^{-1}]$

(B) $[FTV^{-1}]$

(C) $[FT^2\,V]$

(D) $[FVT^{-1}]$

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Q52. MCQ (Single Correct Answer)

Open solution

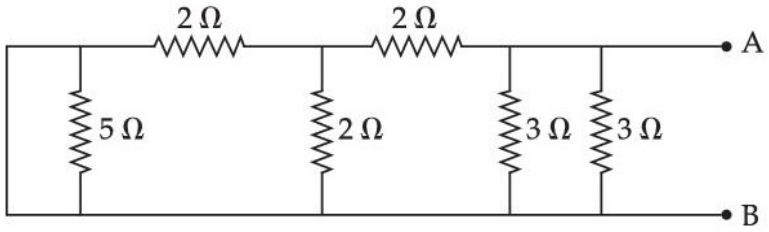
The magnetic field vector of an electromagnetic wave is given by $B = B_0 \frac{\hat{i}+\hat{j}}{\sqrt{2}} \cos(kz - \omega t)$; where $\hat{i}, \hat{j}$ represents unit vector along x and y-axis respectively. At t = 0s, two electric charges q_1 of 4π coulomb and q_2 of 2π coulomb located at $\left(0, 0, \frac{\pi}{k}\right)$ and $\left(0, 0, \frac{3\pi}{k}\right)$, respectively, have the same velocity of 0.5 c $\hat{i}$, (where c is the velocity of light). The ratio of the force acting on charge q_1 to q_2 is :-

(A) $2\sqrt{2} : 1$ (B) $1 : \sqrt{2}$ (C) $2 : 1$ (D) $\sqrt{2} : 1$

Q53. MCQ (Single Correct Answer)

Open solution

The equivalent resistance of the given circuit between the terminals A and B is :



(A) 0Ω (B) 3Ω (C) $\frac{9}{2}\Omega$ (D) 1Ω

Q54. MCQ (Single Correct Answer)

Open solution

Choose the incorrect statement :

(1) The electric lines of force entering into a Gaussian surface provide negative flux.

(2) A charge 'q' is placed at the centre of a cube. The flux through all the faces will be the same.

(3) In a uniform electric field net flux through a closed Gaussian surface containing no net charge, is zero.

(4) When electric field is parallel to a Gaussian surface, it provides a finite non-zero flux.

Choose the most appropriate answer from the options given below

(A) (3) and (4) only (B) (2) and (4) only (C) (4) only (D) (1) and (3) only

Q55. MCQ (Single Correct Answer)

Open solution

A mixture of hydrogen and oxygen has volume 500 cm^3, temperature 300 K, pressure 400 kPa and mass 0.76 g. The ratio of masses of oxygen to hydrogen will be :-

(A) 3 : 8 (B) 3 : 16 (C) 16 : 3 (D) 8 : 3

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Q56. MCQ (Single Correct Answer)

Open solution

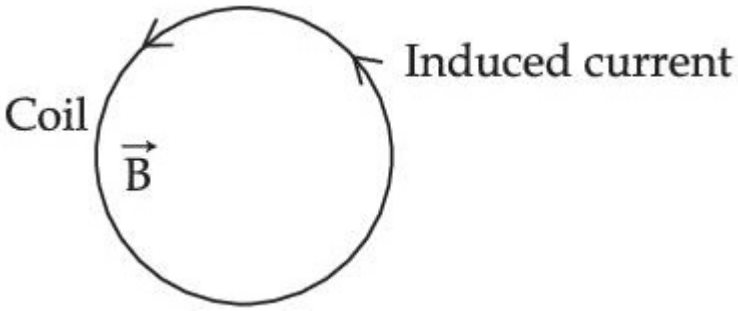
A block moving horizontally on a smooth surface with a speed of 40 m/s splits into two parts with masses in the ratio of 1 : 2. If the smaller part moves at 60 m/s in the same direction, then the fractional change in kinetic energy is :-

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{8}$ (D) $\frac{1}{4}$

Q57. MCQ (Single Correct Answer)

Open solution

A coil is placed in a magnetic field $\overrightarrow{B}$ as shown below :



A current is induced in the coil because $\overrightarrow{B}$ is :

(A) Outward and decreasing with time

(B) Parallel to the plane of coil and decreasing with time

(C) Outward and increasing with time

(D) Parallel to the plane of coil and increasing with time

Q58. MCQ (Single Correct Answer)

Open solution

For a body executing S.H.M. :

(1) Potential energy is always equal to its K.E.

(2) Average potential and kinetic energy over any given time interval are always equal.

(3) Sum of the kinetic and potential energy at any point of time is constant.

(4) Average K.E. in one time period is equal to average potential energy in one time period.

Choose the most appropriate option from the options given below :

(A) (3) and (4) (B) only (3) (C) (2) and (3) (D) only (2)

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Q59. MCQ (Single Correct Answer)

Open solution

Statement - I :

To get a steady dc output from the pulsating voltage received from a full wave rectifier we can connect a capacitor across the output parallel to the load R_L.

Statement - II :

To get a steady dc output from the pulsating voltage received from a full wave rectifier we can connect an inductor in series with R_L.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

Q60. MCQ (Single Correct Answer)

Open solution

If R_E be the radius of Earth, then the ratio between the acceleration due to gravity at a depth 'r' below and a height 'r' above the earth surface is : (Given : $r < R_E$)

(A) $1-\frac{r}{R_E}-\frac{r^2}{R_E^2}-\frac{r^3}{R_E^3}$

(B) $1+\frac{r}{R_E}+\frac{r^2}{R_E^2}+\frac{r^3}{R_E^3}$

(C) $1+\frac{r}{R_E}-\frac{r^2}{R_E^2}+\frac{r^3}{R_E^3}$

(D) $1+\frac{r}{R_E}-\frac{r^2}{R_E^2}-\frac{r^3}{R_E^3}$

Q61. Numerical

Open solution

A parallel plate capacitor of capacitance 200 μF is connected to a battery of 200 V. A dielectric slab of dielectric constant 2 is now inserted into the space between plates of capacitor while the battery remain connected. The change in the electrostatic energy in the capacitor will be _____J.

Q62. Numerical

Open solution

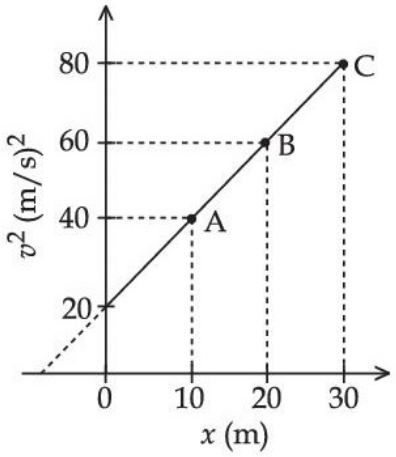
A long solenoid with 1000 turns/m has a core material with relative permeability 500 and volume 10^3 cm^3. If the core material is replaced by another material having relative permeability of 750 with same volume maintaining same current of 0.75 A in the solenoid, the fractional change in the magnetic moment of the core would be approximately $\left(\frac{x}{499}\right)$. Find the value of x.

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Q63. Numerical

A particle is moving with constant acceleration 'a'. Following graph shows v^2 versus x(displacement) plot. The acceleration of the particle is _____ m/s^2.

Open solution



Q64. Numerical

In a Young's double slit experiment, the slits are separated by 0.3 mm and the screen is 1.5 m away from the plane of slits. Distance between fourth bright fringes on both sides of central bright is 2.4 cm. The frequency of light used is _____ $\times 10^{14}$ Hz.

Open solution

Q65. Numerical

The diameter of a spherical bob is measured using a vernier callipers. 9 divisions of the main scale, in the vernier callipers, are equal to 10 divisions of vernier scale. One main scale division is 1 mm. The main scale reading is 10 mm and 8th division of vernier scale was found to coincide exactly with one of the main scale division. If the given vernier callipers has positive zero error of 0.04 cm, then the radius of the bob is _____ $\times 10^{-2}$ cm.

Open solution

Q66. Numerical

A sample of gas with $\gamma = 1.5$ is taken through an adiabatic process in which the volume is compressed from 1200 cm^3 to 300 cm^3. If the initial pressure is 200 kPa. The absolute value of the workdone by the gas in the process = _____ J.

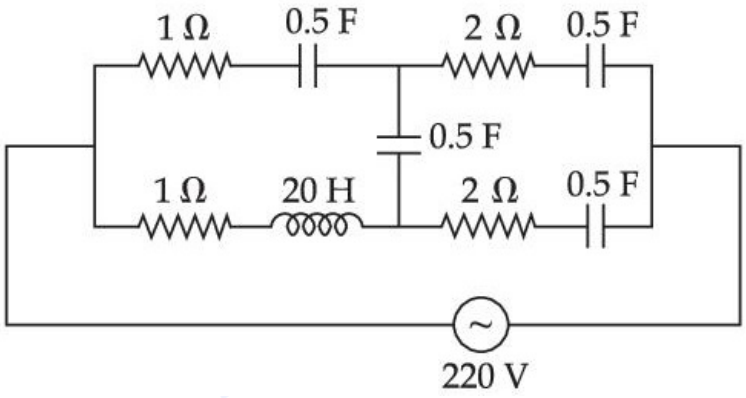
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Q67. Numerical

At very high frequencies, the effective impedance of the given circuit will be _____ Ω.

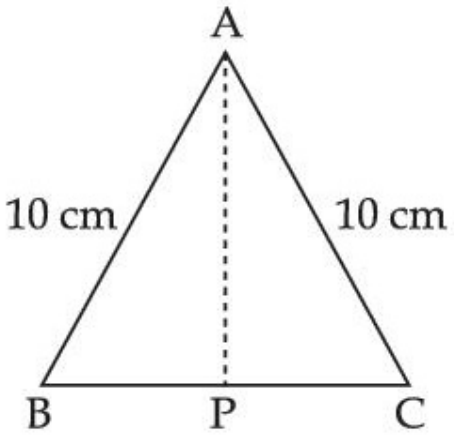
Open solution



Q68. Numerical

Cross-section view of a prism is the equilateral triangle ABC in the figure. The minimum deviation is observed using this prism when the angle of incidence is equal to the prism angle. The time taken by light to travel from P (midpoint of BC) to A is _____ $\times 10^{-10}$ s. (Given, speed of light in vacuum = 3×10^8 m/s and $\cos 30^\circ = \frac{\sqrt{3}}{2}$)

Open solution



Q69. Numerical

A resistor dissipates 192 J of energy in 1s when a current of 4A is passed through it. Now, when the current is doubled, the amount of thermal energy dissipated in 5s in _____ J.

Open solution

Out of syllabus 21 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q70. MCQ (Single Correct Answer) · **Chemistry**

The deposition of X and Y on ground surfaces is referred as wet and dry depositions, respectively. X and Y are :

(A) X = Ammonium salts, Y = CO_2

(B) X = SO_2, Y = Ammonium salts

(C) X = Ammonium salts, Y = SO_2

(D) X = CO_2, Y = SO_2

Open solution

Q71. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Lithium salts are hydrated.

Reason (R) : Lithium has higher polarising power than other alkali metal group members.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(B) (A) is correct but (R) is not correct.

(C) (A) is not correct but (R) is correct.

(D) Both (A) and (R) are correct and (R) is the correct explanation of (A)

Open solution

Q72. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following statements is incorrect?

(A) Atomic hydrogen is produced when H_2 molecules at a high temperature are irradiated with UV radiation.

(B) At around 2000 K, the dissociation of dihydrogen into its atoms is nearly 8.1%

(C) Bond dissociation enthalpy of H_2 is highest among diatomic gaseous molecules which contain a single bond.

(D) Dihydrogen is produced on reacting zinc with HCl as well as $NaOH_{(aq)}$.

Open solution

Q73. MCQ (Single Correct Answer) · **Chemistry**

Which among the following is not a polyester?

(A) Novolac

(B) PHBV

(C) Dacron

(D) Glyptal

Open solution

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Q74. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following correctly represents the order of stability of oxides, X_2O; (X = halogen)?

(A) Br > Cl > I (B) Br > I > Cl (C) Cl > I > Br (D) I > Cl > Br

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

The number of S = O bonds present in sulphurous acid, peroxodisulphuric acid and pyrosulphuric acid, respectively are :

(A) 2, 3 and 4 (B) 1, 4 and 3 (C) 2, 4 and 3 (D) 1, 4 and 4

Open solution

Q76. Numerical · **Chemistry**

CH_4 is adsorbed on 1g charcoal at 0°C following the Freundlich adsorption isotherm. 10.0 mL of CH_4 is adsorbed at 100 mm of Hg, whereas 15.0 mL is adsorbed at 200 mm of Hg. The volume of CH_4 adsorbed at 300 mm of Hg is 10^x mL. The value of x is _____ $\times 10^{-2}$.

(Nearest integer) [Use $\log_{10}2 = 0.3010$, $\log_{10}3 = 0.4771$]

Open solution

Q77. Numerical · **Chemistry**

The empirical formula for a compound with a cubic close packed arrangement of anions and with cations occupying all the octahedral sites in A_xB. The value of x is _____. (Integer answer)

Open solution

Q78. Numerical · **Chemistry**

In the electrolytic refining of blister copper, the total number of main impurities, from the following, removed as anode mud is _____

Pb, Sb, Se, Te, Ru, Ag, Au and Pt

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors mutually perpendicular to each other and have same magnitude. If a vector $\vec{r}$ satisfies

$\vec{a} \times \{(\vec{r} - \vec{b}) \times \vec{a}\} + \vec{b} \times \{(\vec{r} - \vec{c}) \times \vec{b}\} + \vec{c} \times \{(\vec{r} - \vec{a}) \times \vec{c}\} = \vec{0}$, then $\vec{r}$ is equal to :

(A) $\frac{1}{3}(\vec{a} + \vec{b} + \vec{c})$ (B) $\frac{1}{3}(2\vec{a} + \vec{b} - \vec{c})$

(C) $\frac{1}{2}(\vec{a} + \vec{b} + \vec{c})$ (D) $\frac{1}{2}(\vec{a} + \vec{b} + 2\vec{c})$

Open solution

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Q80. MCQ (Single Correct Answer) · **Mathematics**

The distance of the point $(-1, 2, -2)$ from the line of intersection of the planes $2x + 3y + 2z = 0$ and $x - 2y + z = 0$ is :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{5}{2}$ (C) $\frac{\sqrt{42}}{2}$ (D) $\frac{\sqrt{34}}{2}$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

Negation of the statement $(p \vee r) \Rightarrow (q \vee r)$ is :

(A) $p \wedge \sim q \wedge \sim r$ (B) $\sim p \wedge q \wedge \sim r$

(C) $\sim p \wedge q \wedge r$ (D) $p \wedge q \wedge r$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

An angle of intersection of the curves, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $x^2 + y^2 = ab$, $a > b$, is :

(A) $\tan^{-1}\left(\frac{a+b}{\sqrt{ab}}\right)$ (B) $\tan^{-1}\left(\frac{a-b}{2\sqrt{ab}}\right)$

(C) $\tan^{-1}\left(\frac{a-b}{\sqrt{ab}}\right)$ (D) $\tan^{-1}\left(2\sqrt{ab}\right)$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

The number of solutions of the equation $32^{\tan^2 x} + 32^{\sec^2 x} = 81,\ 0 \le x \le \frac{\pi}{4}$ is :

(A) 3 (B) 1 (C) 0 (D) 2

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

Let f be any continuous function on [0, 2] and twice differentiable on (0, 2). If f(0) = 0, f(1) = 1 and f(2) = 2, then

(A) $f''(x) = 0$ for all $x \in (0, 2)$ (B) $f''(x) = 0$ for some $x \in (0, 2)$

(C) $f'(x) = 0$ for some $x \in [0, 2]$ (D) $f''(x) > 0$ for all $x \in (0, 2)$

Open solution

Q85. Numerical · **Mathematics**

Suppose, the line $\frac{x-2}{\alpha} = \frac{y-2}{-5} = \frac{z+2}{2}$ lies on the plane $x + 3y - 2z + \beta = 0$. Then $(\alpha + \beta)$ is equal to _____.

Open solution

Q86. Numerical · **Mathematics**

A tangent line L is drawn at the point $(2, -4)$ on the parabola $y^2 = 8x$. If the line L is also tangent to the circle $x^2 + y^2 = a$, then 'a' is equal to _____.

Open solution

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Q87. Numerical · **Mathematics**

If $S = \frac{7}{5} + \frac{9}{5^2} + \frac{13}{5^3} + \frac{19}{5^4} + \ldots$, then 160 S is equal to _____.

Open solution

Q88. Numerical · **Mathematics**

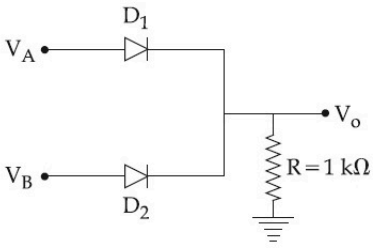
Let B be the centre of the circle $x^2 + y^2 - 2x + 4y + 1 = 0$. Let the tangents at two points P and Q on the circle intersect at the point A(3, 1). Then $8.\left(\frac{area\ \Delta APQ}{area\ \Delta BPQ}\right)$ is equal to _____.

Open solution

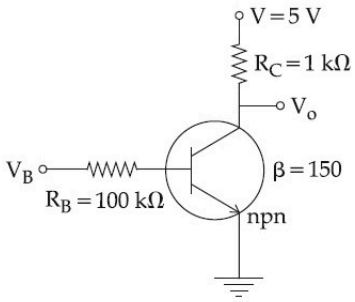
Q89. MCQ (Single Correct Answer) · **Physics**

If V_A and V_B are the input voltages (either 5V or 0V) and V_0 is the output voltage then the two gates represented in the following circuit (A) and (B) are :-

Open solution



(A)



(B)

(A) AND and OR Gate

(B) OR and NOT Gate

(C) NAND and NOR Gate

(D) AND and NOT Gate

Q90. Numerical · **Physics**

A bandwidth of 6 MHz is available for A.M. transmission. If the maximum audio signal frequency used for modulating the carrier wave is not to exceed 6 kHz, the number of stations that can be broadcasted within this band simultaneously without interfering with each other will be _____.

Open solution

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Answer Key

Chemistry

1. **D**	2. **C**	3. **D**	4. *Scan QR*	5. **A**
6. **C**	7. **B**	8. **C**	9. **D**	10. **B**
11. **B**	12. **D**	13. **A**	14. **B**	15. **13**
16. **0**	17. **6021**	18. **47**	19. **13**	20. **0**
21. **7**				

Mathematics

22. **B**	23. **C**	24. **A**	25. **B**	26. **D**
27. **C**	28. **A**	29. **B**	30. **B**	31. **D**
32. **C**	33. **C**	34. **B**	35. **C**	36. **315**
37. **5143**	38. **3**	39. **8**	40. **26**	41. **22**

Physics

42. **B**	43. **D**	44. **C**	45. **B**	46. **C**
47. **A**	48. **C**	49. **D**	50. **D**	51. **B**
52. **C**	53. **D**	54. **C**	55. **C**	56. **C**
57. **A**	58. **A**	59. **D**	60. **D**	61. **4**
62. **250**	63. **1**	64. **5**	65. **52**	66. **480**
67. **2**	68. **5**	69. **3840**		

Out of syllabus

70. **C**	71. **A**	72. **B**	73. **A**	74. **D**
75. **D**	76. **128**	77. **1**	78. **6**	79. **C**
80. **D**	81. **A**	82. **C**	83. **B**	84. **B**
85. **7**	86. **2**	87. **305**	88. **18**	89. **B**
90. **500**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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